

Complex Analytic Geometry

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Abstract

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1 Local Theory of Analytic Varieties

1.1 The Structure Sheaf of $\mathbb{C}^n$

Definition 1.1.1 (The Structure Sheaf of $\mathbb{C}^n$)

Define the structure sheaf $\mathcal{O}_{\mathbb{C}^n} : \mathbf{Open}(\mathbb{C}^n) \rightarrow \mathbf{Ring}$ as follows.

- For each open set $U \subseteq \mathbb{C}^n$, define

$$\mathcal{O}_{\mathbb{C}^n}(U) = \{f : U \rightarrow \mathbb{C} \mid f \text{ is holomorphic}\}$$

- For each inclusion of open sets $V \subseteq U$, define the function $\mathcal{O}_{\mathbb{C}^n}(U) \rightarrow \mathcal{O}_{\mathbb{C}^n}(V)$ to be the restriction function $f \mapsto f|_V$.

Proposition 1.1.2

The following are true regarding the structure sheaf $\mathcal{O}_{\mathbb{C}^n}$.

- $\mathcal{O}_{\mathbb{C}^n}$ is a sheaf.
- If U is connected, then $\mathcal{O}_{\mathbb{C}^n}(U)$ is an integral domain.

Proposition 1.1.3

Let $p \in \mathbb{C}^n$. The following are true regarding the stalk $\mathcal{O}_{\mathbb{C}^n,p}$.

- There is a ring isomorphism $\mathcal{O}_{\mathbb{C}^n,p} \cong \{f \in \mathbb{C}[[z_1, \dots, z_n]] \mid f \text{ is convergent}\}$ for any $p \in \mathbb{C}^n$.
- $\mathcal{O}_{\mathbb{C}^n,p}$ is Noetherian.
- $\mathcal{O}_{\mathbb{C}^n,p}$ is a regular local ring (and hence a UFD and normal).
- $\dim(\mathcal{O}_{\mathbb{C}^n,p}) = n$.

Theorem 1.1.4 (Oka's Coherence Theorem)

$\mathcal{O}_{\mathbb{C}^n}$ is a coherent sheaf of rings.

1.2 Complex Analytic Varieties

Definition 1.2.1 (The Vanishing Locus)

Let $U \subseteq \mathbb{C}^n$ be open. Let $S \subseteq \mathcal{O}_{\mathbb{C}^n}(U)$. Define the vanishing locus of S to be

$$\mathbb{V}(S) = \{z \in U \mid f(z) = 0 \text{ for all } f \in S\}$$

Definition 1.2.2 (Complex Analytic Varieties)

Let $Z \subseteq \mathbb{C}^n$. We say that Z is a complex analytic variety if there exists an open set $Z \subseteq U \subseteq \mathbb{C}^n$ and $f_1, \dots, f_k \in \mathcal{O}_{\mathbb{C}^n}(U)$ such that

$$Z = \mathbb{V}(f_1, \dots, f_k)$$

We use a subset of $\mathbb{C}^n$ in the definition because in $\mathbb{C}^n$, locally defined functions cannot in general be defined globally (compare it to the real case, where they have partitions of unity and bump functions).

Indeed, for smooth functions on $\mathbb{R}^n$, a function on an open set U can be modified (via bump functions) to produce a global function that agrees with it on a smaller open subset. You can always find a global function that captures the local behavior near any point.

For holomorphic functions on $\mathbb{C}^n$: no such modification is possible. A holomorphic function cannot be cut off or modified while remaining holomorphic, so local holomorphic data is genuinely trapped in its domain.

Definition 1.2.3 (The Structure Sheaf) Let $U \subseteq \mathbb{C}^n$ be an open subset. Let $f_1, \dots, f_k \in \mathcal{A}_{\mathbb{C}^n}(U)$ be analytic functions on U . Let $Z = \mathbb{V}(f_1, \dots, f_k)$ be the vanishing locus of $f_1, \dots, f_k$. Define the

structure sheaf

$$\mathcal{O}_Z : \text{Open}(Z) \rightarrow \mathbf{CRing}$$

as follows.

- For each open set $U \subseteq Z$, define

$$\mathcal{O}_Z(U) = \frac{\mathcal{O}_{\mathbb{C}^n}(U)}{(f_1, \dots, f_k)}$$

- For each inclusion $U \subseteq V \subseteq Z$ of open sets, define

$$\mathcal{O}_Z(V) \rightarrow \mathcal{O}_Z(U)$$

to be the induced map of the restriction $\mathcal{O}_{\mathbb{C}^n}(V) \rightarrow \mathcal{O}_{\mathbb{C}^n}(U)$.

1.3 Ruckert's Nullstellensatz

Definition 1.3.1 (Equivalence Class of Pairs of Subsets)

Let $p \in \mathbb{C}^n$. Let A, B, U, V be open sets such that $p \in A \subseteq U$ and $p \in B \subseteq V$. We say that (U, A) and (V, B) are equivalent if there exists an open subset $W \subseteq U \cap V$ such that $A \cap W = B \cap W$.

Definition 1.3.2 (The Vanishing Set of a Germ)

Let $[(f, U)] \in \mathcal{O}_{\mathbb{C}^n, 0}$. Define the vanishing set of f to be

$$\mathbb{V}_0(f) = [(U, \mathbb{V}(f))]$$

Definition 1.3.3 (The Vanishing Set of an Ideal)

Let $I \subseteq \mathcal{O}_{\mathbb{C}^n, 0}$ be an ideal. Suppose that $I = [(f_1, U_1), \dots, (f_k, U_k)]$. Define the vanishing set of I to be

$$\mathbb{V}_0(I) = \bigcap_{i=1}^k \mathbb{V}_0(f_i)$$

Definition 1.3.4 (The Ideal of a Subset)

Let $X \subseteq \mathbb{C}^n$ be a subset. Define the ideal of X to be

$$\mathbb{I}_0(X) = \{f \in \mathcal{O}_{\mathbb{C}^n, 0} \mid \mathbb{V}_0(f) \supseteq X\}$$

Theorem 1.3.5 (Ruckert's Nullstellensatz)

Let $I \subseteq \mathcal{O}_{\mathbb{C}^n, 0}$ be an ideal. Then we have

$$\sqrt{I} = \mathbb{I}_0(\mathbb{V}_0(I))$$

1.4 Complex Analytic Subvarieties of Manifolds

Definition 1.4.1 (Complex Analytic Subvarieties)

Let M be a complex manifold. Let $Z \subseteq M$ be a subset. We say that Z is a complex analytic subvariety of M if for each $p \in Z$, there exists a chart (U, ϕ) of M such that $\phi(Z \cap U)$ is a complex analytic variety.

In particular, complex analytic subvarieties of $\mathbb{C}^n$ are just complex analytic varieties.

Theorem 1.4.2 (Cartan's (Coherence) Theorem) Let M be a complex manifold and let A be an analytic subset of M . Then $\mathcal{I}_A$ is a coherent sheaf of $\mathcal{O}_M$ -modules.

Theorem 1.4.3 (Riemann–Stein theorem)

1.5 Regular and Singular Points

Definition 1.5.1 (Regular and Singular Points) Let M be a complex manifold and let A be an analytic subset of M . We say that $x \in A$ is a regular point if there exists some open neighbourhood U of x such that $A \cap U$ is a complex submanifold of M . Otherwise, x is said to be singular. Denote

$$A_{\text{reg}} = \{x \in A \mid x \text{ is a regular point}\} \quad \text{and} \quad A_{\text{sing}} = \{x \in A \mid x \text{ is a singular point}\}$$

Theorem 1.5.2 Let M be a complex manifold and let A be an analytic subset of M . Then A_{sing} is an analytic subset of A .

2 Complex Analytic Spaces

Definition 2.0.1 (Complex Analytic Spaces)

Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is a complex analytic space if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is a complex analytic variety.

Lemma 2.0.2

Let $(X, \mathcal{O}_X)$ be a complex analytic space. Then X is a complex manifold if and only if for each $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is an isomorphism

$$(U, \mathcal{O}_X|_U) \cong (V, \mathcal{O}_{\mathbb{C}^n}|_V)$$

for some open subset $V \subseteq \mathbb{C}^n$.

Theorem 2.0.3 Let X be a complex space. Then X_{reg} is dense and open in X . Moreover, X_{reg} consists of a disjoint union of complex manifolds.

Theorem 2.0.4 Let X be an irreducible complex space. Then every non-constant holomorphic function $f : X \rightarrow \mathbb{C}$ is an open map.

Corollary 2.0.5

Let X be an irreducible compact complex space. Then we have $\mathcal{O}_X(X) \cong \mathbb{C}$.

Definition 2.0.6 (Complex Analytic Subvarieties)

Let $(X, \mathcal{O}_X)$ be a complex analytic space. Let $Z \subseteq X$ be subset. We say that Z is a complex analytic subvariety if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ and $f_1, \dots, f_k \in \mathcal{O}_X(U)$ such that

$$Z \cap U = \mathbb{V}_U(f_1, \dots, f_k)$$

This definition generalizes complex analytic subvarieties of complex manifolds.

3 Divisors and Line Bundles

3.1 Divisors

Definition 3.1.1 (Divisors on a Complex Manifold)

Let X be a complex manifold. A divisor on X is a formal linear combination

$$D = \sum_{\substack{V \text{ is an analytic variety of } X \\ \text{of dimension } \dim(X)-1}} n_V \cdot V$$

where $n_V \in \mathbb{Z}$, and such that for each $p \in X$, there exists an open neighbourhood U of p such that U meets finitely many V in the summand.

The locally finite condition here is required because in Algebraic Geometry, Noetherianity is assumed, otherwise they are the same.

Definition 3.1.2 (Group of Divisors)

Let X be a complex manifold. Define the group of divisors of X to be the set

$$\text{Div}(X) = \{D \mid D \text{ is a divisor on } X\}$$

together with addition given as a subgroup of $\prod_V \mathbb{Z} \cdot V$.

Definition 3.1.3 (Effective Divisors)

Let X be a complex manifold. Let $D = \sum_V n_V \cdot V \in \text{Div}(X)$ be a divisor. We say that D is effective if $n_V \geq 0$ for all V .

Definition 3.1.4 (Divisors of Meromorphic Functions)

Let X be a complex manifold. Let $f \in \mathcal{M}_X^*$ be a meromorphic function. Define the divisor of f to be

$$\text{div}(f) = \sum_V \nu_V(f) \cdot V$$

where ν_V is the valuation of $\mathcal{O}_{X,V}$.

Definition 3.1.5 (Cartier Divisors on a Complex Manifold)

Let X be a complex manifold. Define the group of divisors of X to be

$$\text{CDiv}(X) = H^0(X, \mathcal{M}_X^* / \mathcal{O}_X^*) = \Gamma(X, \mathcal{M}_X^* / \mathcal{O}_X^*)$$

Proposition 3.1.6

Let X be a complex manifold. Then there is a natural isomorphism

$$\text{Div}(X) \cong \text{CDiv}(X)$$

3.2 Holomorphic Line Bundles

Proposition 3.2.1

Let X be a complex manifold. Then the following diagram is commutative:

$$\begin{array}{ccc} \text{CDiv}(X) & \longrightarrow & \text{Pic}(X) \\ \cong \downarrow & & \downarrow \cong \\ H^0(X, \mathcal{M}_X^* / \mathcal{O}_X^*) & \longrightarrow & H^1(X, \mathcal{O}_X^*) \end{array}$$

where the bottom horizontal map is induced by the long exact sequence of the short exact sequence $0 \rightarrow \mathcal{O}_X^* \rightarrow \mathcal{M}_X^* \rightarrow \frac{\mathcal{M}_X^*}{\mathcal{O}_X^*} \rightarrow 0$.

By Serre's correspondence, holomorphic line bundles correspond to locally free sheaves of $\mathcal{O}_M$ -modules of rank 1, and recall isomorphism classes of these objects form a group structure with the tensor product, and it is group isomorphic to $H^1(M, \mathcal{O}_M^*)$.

Definition 3.2.2 (The Exponential Sequence)

Let M be a complex manifold. Define the exponential sequence of M to be the short exact sequence of sheaves of $\mathcal{O}_M$ -modules given by

$$0 \longrightarrow \underline{\mathbb{Z}} \longrightarrow \mathcal{O}_M \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_M^* \longrightarrow 0$$

Proposition 3.2.3

Let M be a complex manifold. Let L be a holomorphic line bundle on M . Then the first chern class of L is given by the image of L in the homomorphism

$$\text{Pic}(M) \cong H^1(M, \mathcal{O}_M^*) \rightarrow H^2(M, \underline{\mathbb{Z}}) \cong H^2(M, \mathbb{Z})$$

induced by the long exact sequence of the exponential sequence.

3.3 Kodaira Dimension

Definition 3.3.1

Let M be a complex manifold. Let L be a holomorphic line bundle on M . Define $R(M, L)$ to be

$$R(M, L) = \bigoplus_{m=0}^{\infty} H^0(M, L^{\otimes m})$$

Definition 3.3.2 (The Canonical Ring)

Let M be a complex manifold. Define the canonical ring of M to be

$$R(M) = R(M, \mathcal{M}_M) = \bigoplus_{m=0}^{\infty} H^0(M, \mathcal{M}_M^{\otimes m})$$

Lemma 3.3.3

Let M be a complex manifold. If M is connected, then $R(M)$ is an integral domain.

Definition 3.3.4 (Kodaira Dimension)

Let M be a connected complex manifold. Define the Kodaira dimension of M to be

$$\text{kod}(M) = \begin{cases} -\infty & \text{if } R(M) = \mathbb{C} \\ \text{trdeg}_{\mathbb{C}}(R(M)) - 1 & \text{otherwise} \end{cases}$$

4 Stein Manifolds